

PROJECT REPORT
ON
SOCIAL MEDIA POST ENGAGEMENT
PREDICTION USING MACHINE LEARNING

Bring Your Own Project (BYOP) Submission
for the course
Fundamentals in AI and ML

Submitted By:

Name: AAKARSHA PARASHAR

Registration No.: 25BAI10492

Programme / Branch: CSE AIML

Submitted To:

Faculty Name: Prakash N B

VIT

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CERTIFICATE

This is to certify that the project report entitled “Social Media Post Engagement Prediction Using Machine Learning” is a Bonafide work carried out by the student in partial fulfillment of the requirements for the course Fundamentals in AI and ML during the academic year 2025–2026.

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1. ABSTRACT

Social media has become a major platform for communication, branding, promotion, and business growth. However, one of the most common challenges faced by content creators, influencers, and businesses is predicting how well a post will perform before publishing it. Engagement such as likes, comments, and shares depends on multiple factors, and poor posting decisions often lead to reduced visibility and low audience interaction.

This project, titled EngageAI, aims to solve this problem by building a machine learningbased prediction system that estimates the likely engagement level of a social media post. The model uses features such as post type, caption length, number of hashtags, posting day, posting hour, and follower count to classify a post into one of three categories: High, Medium, or Low engagement.

The project applies data preprocessing, feature engineering, and classification techniques using Python and Scikit-learn. A simple Streamlit-based user interface is also used to make the system interactive and easy to use. The final system can help users make more informed and data-driven decisions while planning social media content.

2. INTRODUCTION

In today's digital era, social media plays an essential role in communication, marketing, and personal branding. Platforms such as Instagram, Facebook, LinkedIn, and X (formerly Twitter) are widely used by individuals, businesses, and organizations to reach audiences and promote content.

One of the biggest concerns for social media users is engagement, which includes likes, comments, shares, and audience interaction. Higher engagement generally indicates that a post is performing well and reaching more people. However, predicting engagement is not easy because it depends on several variables such as the type of content, posting time, use of hashtags, and audience size.

Many creators still post content based on assumptions rather than data-driven insights. This leads to inconsistent results and missed opportunities. Artificial Intelligence and Machine Learning can help solve this issue by identifying patterns in past post performance and using those patterns to predict future engagement.

This project focuses on developing a machine learning model that can estimate the likely performance of a social media post before it is published.

3. PROBLEM STATEMENT

Social media creators, businesses, and marketers often do not know which type of content is likely to perform well. Engagement is influenced by multiple factors, but without proper analysis, posting strategies remain inconsistent and ineffective.

The problem addressed in this project is the lack of a predictive system that can estimate the expected engagement level of a social media post using its content-related and posting-related characteristics.

4. WHY THIS PROBLEM MATTERS

This problem is relevant in real life because social media has become a major tool for business promotion, personal branding, communication, and audience building. Small businesses and creators often rely heavily on social media visibility, but many do not have access to advanced analytics tools or data science support.

A prediction system that estimates engagement before publishing can help users make smarter content decisions, improve posting consistency, and reduce trial-and-error in digital strategy. This makes the problem practical, modern, and highly relevant to AI/ML applications.

5. OBJECTIVES

The main objectives of this project are:

- To study factors affecting social media post engagement
- To analyze historical engagement data
- To create an engagement score using likes, comments, and shares
- To classify posts into High, Medium, and Low engagement levels
- To build a machine learning model for engagement prediction
- To create a simple prediction interface for end users

6. SCOPE OF THE PROJECT

The scope of this project includes social media post-level structured data analysis, engagement category prediction using machine learning, and a simple deployable prediction interface. The system focuses on input variables such as post type, caption length, posting time, hashtags count, and follower count.

The project does not include image or video content analysis, live API integration from social media platforms, or real-time audience behavior tracking. This keeps the work focused, achievable, and aligned with the course objectives.

7. PROPOSED SOLUTION

The proposed solution is a machine learning-based prediction system that takes relevant post details as input and predicts the likely engagement level of the post.

The solution includes collecting or using a dataset containing post performance data, cleaning and preprocessing the dataset, creating an engagement score from likes, comments, and shares, categorizing posts into High, Medium, and Low engagement, training a machine learning classification model, saving the trained model for prediction, and building a simple user interface using Streamlit.

This solution allows users to test post configurations before publishing them and use data-driven insights to plan content better.

8. DATASET DESCRIPTION

The dataset used in this project contains structured data related to social media post performance.

Features used in the project include:

- post_type → Type of post (Image, Reel, Video, Carousel)
- caption_length → Number of words in the caption
- hashtags_count → Number of hashtags used
- post_day → Day of posting
- post_hour → Hour of posting
- likes → Number of likes received
- comments → Number of comments received
- shares → Number of shares received
- follower_count → Number of followers at the time of posting

A new target variable called performance_category was created based on the total engagement score.

9. METHODOLOGY

The project follows the standard machine learning workflow.

Step 1: Data Collection

A dataset containing social media post-related variables was collected and used as the base for analysis and model building.

Step 2: Data Cleaning

The dataset was cleaned by removing missing values, removing duplicate records, and checking column consistency.

Step 3: Feature Engineering

A new column called engagement_score was created: engagement_score
= likes + comments + shares

This score was then divided into Low, Medium, and High categories based on percentile thresholds.

Step 4: Encoding

Categorical variables such as post type and posting day were converted into numerical format using Label Encoding.

Step 5: Train-Test Split

The dataset was split into 80% training data and 20% testing data.

Step 6: Model Training

A Random Forest Classifier was used to train the model.

Step 7: Model Evaluation

The model was evaluated using Accuracy, Classification Report, and Confusion Matrix.

Step 8: Deployment

The trained model was integrated into a Streamlit web application.

10. MACHINE LEARNING MODEL USED

The model used in this project is Random Forest Classifier.

This model was chosen because it works well on structured/tabular data, handles multiple features effectively, reduces overfitting compared to a single decision tree, provides good prediction performance, and is easy to explain in academic context.

Although other models could also be used, Random Forest was selected because it offers a good balance of simplicity and performance.

11. TOOLS AND TECHNOLOGIES USED

Programming Language:

- Python

Libraries:

- Pandas
- NumPy
- Scikit-learn
- Joblib
- Streamlit

Development Tools:

- VS Code / Jupyter Notebook
- GitHub

12. IMPLEMENTATION OVERVIEW

The implementation of the project involved loading the dataset, cleaning and preprocessing the data, creating engagement categories, training the machine learning model, saving the trained model and encoders, and using a Streamlit app for predictions.

The user interface allows the user to enter post details and receive an engagement prediction instantly. This makes the project not only analytical but also practically usable.

13. KEY DECISIONS MADE DURING THE PROJECT

Several important design choices were made during development.

1. Choosing Classification Instead of Regression

Instead of predicting exact likes or comments, the project predicts engagement levels (High / Medium / Low). This made the output more practical and easier to interpret.

2. Using Structured Features Only

The project focuses only on structured features like hashtags, caption length, posting day, etc. This kept the project manageable and aligned with the course scope.

3. Using Random Forest

Random Forest was chosen due to its strong performance on tabular data and ease of implementation.

4. Using Streamlit

A Streamlit interface was selected because it is simple, lightweight, and suitable for academic demonstration.

14. CHALLENGES FACED

During the development of this project, several challenges were encountered:

- Difficulty in selecting the most relevant input features
- Managing categorical variables and encoding them correctly
- Converting raw engagement data into meaningful target labels
- Ensuring that training and prediction use the same encoders
- Deploying the Streamlit app correctly with all required dependencies

These challenges helped improve practical understanding of ML project development.

15. RESULTS

The final model was able to classify social media posts into engagement categories based on input features.

The system predicts:

- High Engagement
- Medium Engagement
- Low Engagement

This makes the model useful for planning better posting strategies, optimizing content timing, and improving social media performance.

The project demonstrates that machine learning can be used effectively for content strategy decision-making.

16. LEARNING OUTCOMES

This project helped in understanding and applying the following concepts:

- Real-world problem identification
- Supervised machine learning
- Classification modeling
- Feature engineering
- Label encoding
- Model evaluation
- Deployment basics using Streamlit
- Project documentation and version control

It also improved practical confidence in building an end-to-end AI/ML solution.

17. REFLECTION

This project was meaningful because it addressed a real-world issue that is highly relevant in today's digital environment. It showed how AI/ML can be applied to everyday problems, not just complex technical systems.

One of the most valuable lessons from this project was understanding that a good AI/ML solution does not need to be extremely complex. A project becomes impactful when it solves a real problem in a clear, useful, and understandable way.

This BYOP experience helped bridge the gap between theoretical learning and practical implementation.

18. CONCLUSION

The project EngageAI: Social Media Post Engagement Prediction Using Machine Learning successfully demonstrates how machine learning can be applied to social media analytics.

By using historical post data and structured engagement-related features, the system can predict the likely engagement level of a post. This can help users improve their content planning and make better posting decisions.

The project meets the objective of solving a practical real-world problem using AI/ML concepts learned in the course and serves as a meaningful capstone submission.

19. FUTURE SCOPE

This project can be enhanced in the future by:

- Adding caption sentiment analysis
- Analyzing image and video content
- Integrating real-time social media APIs
- Predicting exact likes/comments/shares
- Recommending best posting time and content format
- Adding dashboard analytics for performance monitoring

These improvements can make the project more powerful and industry-oriented.

20. REFERENCES

- Python Documentation
- Scikit-learn Documentation
- Streamlit Documentation
- Kaggle datasets (if used)
- Course notes and class materials